WEB SERVICES & WEB DATA — COURSEWORK 1

MusicMood

A music listening analytics API

FastAPI · SQLite · JWT Auth · Llama 3 via Groq · Railway

5,000+	4	15+	AI
Tracks	Models	Endpoints	Powered

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What does your music say about your mood?

Real Data

5,000 tracks seeded from Kaggle Spotify dataset with pre-computed audio features

Psychology-grounded

Russell's circumplex model maps valence + energy to four mood quadrants

AI Interpretation

Llama 3 interprets pre-computed analytics — not raw data. A genuine AI layer, not a wrapper

Motivated by the intersection of music psychology and software engineering

SYSTEM ARCHITECTURE

UI Layer

HTML/CSS/JS served directly from FastAPI at /ui

Auth Layer

JWT (HS256) · bcrypt password hashing · Protected routes

Data Layer

User · Track · ListeningSession · MoodSnapshot — SQLAlchemy ORM

Analytics Layer

Mood trend · Context breakdown · Genre emotion map · Top tracks

AI Layer

Aggregated analytics → structured prompt → Llama 3 → JSON response with disclaimer

DATA MODELS & ACADEMIC GROUNDING

User

id · username · email
hashed_password · created_at

bcrypt hashing — never plain text

Track

title · artist · genre
valence · energy · danceability
tempo · spotify_id

Spotify audio features from Kaggle dataset

ListeningSession

user_id · track_id
listened_at · context (enum)

Context enum grounded in Schäfer et al. (2013)

MoodSnapshot

computed_mood_score
dominant_emotion · created_at

Derived from Russell circumplex model

API ENDPOINTS

Auth	Tracks (CRUD)	Sessions	Analytics	AI ✦
`POST /auth/register`	`GET /tracks/`	`POST /sessions/`	`GET /analytics/mood-trend/{user_id}`	`POST /ai/interpret`
`POST /auth/login`	`POST /tracks/`	`GET /sessions/{user_id}`	`GET /analytics/context-breakdown/{user_id}`	`GET /ai/recommend-context`
`GET /auth/me`	`PUT /tracks/{id}`	`DELETE /sessions/{id}`	`GET /analytics/genre-emotion-map`	
	`DELETE /tracks/{id}`			

✦ *AI endpoints receive pre-computed analytics from the DB — Llama 3 acts as an interpretation layer, not a chatbot*

LIVE DEMO

1

Homepage

Genre emotion map — live data from DB · 5,000+ tracks · real Spotify features

2

Register + Dashboard

Create account → mood score, AI interpretation, context breakdown, mood trend

3

Swagger /docs

Show endpoint structure → hit GET /analytics/mood-trend live → show raw JSON

4

AI Endpoint

POST /ai/interpret → structured prompt → Llama 3 response with non-clinical disclaimer

VERSION CONTROL & DEPLOYMENT

Branch Strategy

`main`	Production ready — merged via PRs only
`feature/auth`	JWT authentication
`feature/crud`	Track + session endpoints
`feature/analytics`	Analytics computation layer
`feature/ai`	AI interpretation endpoints
`feature/database_models`	SQLAlchemy models and db set up
`feature/sessions`	Listening endpoints
`feature/ui`	HTML

Deployment

Platform

Railway (ASGI/FastAPI compatible)

Config

Procfile with uvicorn ASGI server

Env vars

.env excluded from repo via .gitignore

Lesson

Deploy early — late deployment revealed environment differences

GENERATIVE AI USAGE

Tool used: Claude (Anthropic) — declared and documented throughout

Conceptual Understanding

Used Claude to deeply understand Russell's circumplex model, JWT algorithm tradeoffs (HS256 vs RS256), and the architectural distinction between an LLM wrapper and a genuine AI layer

Architecture Design

Explored alternatives for the AI endpoint design — the decision to compute analytics first then pass structured summaries to Llama 3 emerged from iterative discussion with Claude

Risk Analysis

Used Claude to stress-test the project — identifying risks like overclaiming psychology, thin datasets, and superficial AI use — then designing mitigations into the architecture

Code & Documentation

Assisted with FastAPI patterns, SQLAlchemy relationships, and Pydantic validation. All generated code was understood, reviewed and adapted — not blindly copied

Summary

- ✓ RESTful API with full CRUD, auth, analytics, and AI interpretation
- ✓ Data model grounded in Schäfer (2013) and Russell's (1980) circumplex model
- ✓ AI layer interprets pre-computed analytics — not a wrapper
- ✓ Deployed on Railway with UI served directly from FastAPI
- ✓ GenAI used for conceptual understanding, architecture and risk analysis

Thank you